1. Identity

Document	STAR-PKG-STAR-INTERP-EN · v2.0 · 2026-10-21
Package	star-interp
Version	0.1.0
Capability	star.numerics.lagrange-interpolation
Maturity	RELEASED
Python	⩾3.9
Source commit	94fcf7c
Source	https://github.com/Andrea07072000/S.T.A.R.-/tree/main/packages/star_interp

2. Licence and obligations

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3. What it needs at run time

Nothing beyond the Python standard library: it works offline.

4. Integrity

SHA-256 fingerprints of the files published on this website. Check a downloaded file with:

```
python -c "import hashlib,sys;print(hashlib.sha256(open(sys.argv[1],'rb').read()).hexdigest())" star_interp-0.1.0-py3-none-any.whl
```

FILE	SHA-256
star_interp-0.1.0-py3-none-any.whl	a45b9a02d54585bf650f7d1ab8e6dbf2ed0e0160334921d5bd2806a0a0c38d6e
star_interp.py	11c6ad197ef4621720a7ce1ed8c0464b407f6b9fedcdae92ec7c361063a96e7c
star_interp-0.1.0-source.zip	e732680ec177adac6fcb2865d1e903bcb0d991a0a7349fde81d9c9abd43c6866

5. What it does, and its limits

Value and first derivative of the interpolating polynomial through 2 to 20 tabulated points, in barycentric form; within 7e-15 of exact rational arithmetic on 600 tables, closer to the exact value than SciPy (1.2e-14) and NumPy (7e-11) on the same tables; extrapolation refused.

The accuracy stated is that of evaluating the polynomial, NOT of the interpolation: how well a polynomial through n points represents an orbit depends on the spacing, the degree and where in the table x falls (the central interval is the reliable one), and is the caller's to assess. No extrapolation, no Hermite interpolation (positions with velocities), no splines, scalars only (interpolate each coordinate separately). With many points far from equally spaced the polynomial itself oscillates; that is a property of the method, not checked here.

- `lagrange(xs, ys, x)`: the polynomial of degree len(xs) − 1 through the points, in barycentric form; nodes in any order and any spacing; at a node the tabulated value is returned exactly. Reproduces Meeus Example 3.a.
- `lagrange_derivative(xs, ys, x)`: its first derivative, also at the nodes; `lagrange_weights(xs)`: the weights.
- Measured on 600 tables of 2 to 11 jittered, unsorted nodes against EXACT rational arithmetic (SymPy): value within 7e-15 of the exact polynomial, relative to the largest tabulated value; derivative within 1e-13 (scaled). On the same tables SciPy's barycentric interpolator is within 1.2e-14 of the exact value and a NumPy polynomial fit within 7e-11.
- Every function returns a finite float or raises `ValueError`: fewer than 2 or more than 20 points, lengths that differ, equal nodes, any non-finite or non-numeric element (booleans included), an x outside the range of the nodes (extrapolation is refused), nodes whose weights overflow.

6. Export, use and certification

The package is published and freely available. That usually takes software out of export-control lists, but not every use of it: read the page on space activities and export control before using it in a controlled programme. It is not certified or qualified for flight; the evidence describes what was tested and how.

7. Citation and bill of materials

```
@software{cavazzini_star_interp_0_1_0,
  author  = {Cavazzini, Andrea},
  title   = {{star-interp}: Lagrange interpolation through tabulated points},
  version = {0.1.0},
  year    = {2026},
  url     = {https://andrea07072000.github.io/S.T.A.R.-/software/star_interp/},
  note    = {Apache-2.0. Source: https://github.com/Andrea07072000/S.T.A.R.-, commit 94fcf7c}
}
```

CITATION.cff · BibTeX · SBOM CycloneDX 1.5 · SBOM SPDX 2.3

8. Warranty and liability

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